Provenance of the Electrostatic Material Library

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The library

`statlib_catalog.dat` is an electrostatic material library for FEMM, 94 entries, built 1 October 2026. An electrostatic material is a relative permittivity, ε_x and ε_y (they differ only for the anisotropic 6H-SiC entry, whose c-axis lies along y), and a volume charge density, zero here. Electrostatics is the zero-frequency limit, so each dielectric takes the lowest-frequency value its source gives; the frequency column says which. The solid-insulation, insulating-oil and laminate values are shared with the current-flow library and live in its CSV files.

A convenience entry, "Vacuum", with $\varepsilon_r = 1$ by definition, sits above the folders, as "Air" does in the magnetics library; it needs no source and is not counted or tabulated below. The library is generated by `statlib_src/build_statlib.py`; `statlib_src/catalog.csv` lists every other entry, and the table below is generated from it. `SOURCES.md` records each source document with its SHA-256 hash.

Liquids

The electrostatic solver treats every material as a perfect insulator, and water is not one. Even ultrapure water at 25 °C, with $\sigma = 5.5$ μS/m and $\varepsilon_r = 78.3$ (the current-flow library's entry, from ITU-R P.527-6 and Light et al.), relaxes free charge in $\varepsilon_0\varepsilon_r/\sigma = 0.13$ ms. In a field held longer than that, water behaves as a conductor and its permittivity does not enter. The water entries therefore sit in a folder named "Water (high-frequency limit; DC: use current flow)", and a problem with water in it belongs in the current-flow solver, which models conduction and permittivity together. A library entry carries no comment, so the folder name is the only warning a user sees.

The organic liquids with ε_r above 10 (the alcohols, glycerol, acetone, formamide, nitrobenzene and 1,2-dichloroethane) sit in "Polar Organic Liquids (high-frequency limit)" for the same reason. Polar solvents dissolve and dissociate ionic impurities, so they conduct too, by an amount that depends on purity; no held source gives a value. The low-permittivity organic liquids and the insulating oils are good insulators and can be used as they stand.

Check against the CRC Handbook

The *CRC Handbook of Chemistry and Physics* [14] covers 51 of the entries; the "CRC" column gives its value, table and the library's difference from it (for gases, the difference in $\varepsilon - 1$, marked *). 43 agree within 5%, 5 within 10%, and 3 differ by more. The gases agree within 1% in $\varepsilon - 1$, but CRC's values descend partly from the same NBS Circular 537, so they are not independent. The liquids are an independent check: CRC's are Wohlfarth's Landolt–Börnstein compilation (1991), evaluated at the library's temperature from CRC's fits where it tabulates another; water agrees within 0.3%, and the other liquids within 2% except 2-propanol (-5%) and glycerol (-7%). Its glass values are at 100 MHz, its inorganic-solid values often in the infrared, and its polymer values "typical or average", so differences of several per cent there are expected.

The larger differences: Si_3N_4 (Dorf's 12.7 against CRC's 4.2 for a film; both are far from the 7–8 usually quoted for silicon nitride, and neither is a reliable figure); Ta_2O_5 (Dorf's 28 against 24 for the β phase); nylon 6,6 (Callister's 4.0 against 3.5 for polyamides generally; nylon absorbs water, which raises it). Silicon's CRC value is at 4.2 K and is not comparable.

Entries

Entry	ε_r	Frequency	Src.	CRC	Note
Gases (20 C, 1 atm)					
Helium	1.000065	static	[1]	1.0000645 (6-221, +1%*)	(eps-1) x 10^6 = 65.0
Hydrogen	1.0002538	static	[1]	1.0002532 (6-221, +0%*)	(eps-1) x 10^6 = 253.8
Oxygen	1.0004947	static	[1]	1.0004941 (6-221, +0%*)	(eps-1) x 10^6 = 494.7
Argon	1.0005172	static	[1]	1.0005169 (6-221, +0%*)	(eps-1) x 10^6 = 517.2
Air (dry, CO2-free)	1.0005364	static	[1]	1.000536 (6-221, +0%*)	(eps-1) x 10^6 = 536.4
Nitrogen	1.000548	static	[1]	1.0005474 (6-221, +0%*)	(eps-1) x 10^6 = 548.0
Carbon dioxide	1.000922	static	[1]	1.0009217 (6-221, +0%*)	(eps-1) x 10^6 = 922
Liquids / Water (high-frequency limit; DC: use current flow)					
Water, 20 C	80.37	static	[2]	80.1 (6-199, +0%)	20 C; Section A standard liquid
Water, 25 C	78.54	static	[2]	78.36 (6-15, +0%)	25 C; Section A standard liquid
Water, 50 C	70	static	[2]	69.88 (6-15, +0%)	50 C; from 78.54 at 25 C and the circular's coefficient a = 0.00200 per C, log10(eps) = log10(eps25) - a (t - 25); ITU-R P.527-6 eq. 8 gives 70.3
Liquids / Polar Organic Liquids (high-frequency limit)					
Methanol	33.62	static	[2]	33 (6-199, +2%)	20 C; Section A standard liquid (32.63 at 25 C)
Ethanol	24.3	static	[2]	24.85 (6-199, -2%)	25 C
1-Propanol	20.1	static	[2]	20.52 (6-199, -2%)	25 C
2-Propanol	18.3	static	[2]	19.26 (6-199, -5%)	25 C
Glycerol	42.5	static	[2]	45.64 (6-199, -7%)	25 C
Acetone	20.7	static	[2]	20.49 (6-199, +1%)	25 C
Formamide	109	static	[2]	111 (6-199, -2%)	20 C
1,2-Dichloroethane	10.65	static	[2]	10.42 (6-199, +2%)	20 C; Section A standard liquid (10.36 at 25 C)
Nitrobenzene	35.74	static	[2]	35.6 (6-199, +0%)	20 C; Section A standard liquid (34.82 at 25 C)
Liquids / Low-Permittivity Organic Liquids					
Diethyl ether	4.335	static	[2]	4.2666 (6-199, +2%)	20 C
Benzene	2.2811	static	[2]	2.2825 (6-199, -0%)	20 C; Section A standard liquid (2.274 at 25 C)
Toluene	2.379	static	[2]	2.374 (6-199, +0%)	25 C
Hexane	1.89	static	[2]	1.8865 (6-199, +0%)	20 C
Cyclohexane	2.023	static	[2]	2.0243 (6-199, -0%)	20 C; Section A standard liquid (2.015 at 25 C)
Carbon tetrachloride	2.238	static	[2]	2.2379 (6-199, +0%)	20 C; Section A standard liquid (2.228 at 25 C)
Chlorobenzene	5.708	static	[2]	5.6895 (6-199, +0%)	20 C; Section A standard liquid (5.621 at 25 C)
Liquids / Insulating Oils					
Capacitor oil	2.2	60 Hz	[3]	–	25 C
Pipe cable oil	2.15	60 Hz	[3]	–	25 C
Self-contained cable oil	2.3	60 Hz	[3]	–	25 C
Heavy cable oil	2.23	60 Hz	[3]	–	25 C

Entry	ε_r	Frequency	Src.	CRC	Note
Transformer oil	2.25	60 Hz	[3]	–	25 C
Alkyl benzene	2.1	60 Hz	[3]	–	25 C
Polybutene pipe cable oil	2.14	60 Hz	[3]	–	25 C; dielectric constant given at 1 MHz
Polybutene capacitor oil	2.22	60 Hz	[3]	–	25 C; dielectric constant given at 1 MHz
Silicone fluid	2.7	60 Hz	[3]	–	25 C
Castor oil	3.74	60 Hz	[3]	–	25 C
C8F16O fluorocarbon	1.86	60 Hz	[3]	–	25 C
Solid Insulation / Ceramics, Glass and Stone					
Alumina (Al2O3)	8.5	60 Hz	[3]	–	
Porcelain (mullite)	8.2	60 Hz	[3]	–	
Steatite	5.5	60 Hz	[3]	–	
Magnesium oxide (MgO)	9.65	60 Hz	[3]	9.65 (12-48, +0%)	
Glass (soda lime)	6.25	60 Hz	[3]	6.75 (12-58, -7%)	
Mica	6.9	60 Hz	[3]	6.9 (12-48, +0%)	
Porcelain	6	60 Hz	[5]	–	60 Hz; Dorf Table 55.2 gives 8.2 for mullite porcelain
Fused silica	4	60 Hz	[5]	4.1 (12-58, -2%)	60 Hz
Pyrex 7740 borosilicate glass	4.6	1 MHz	[11]	5 (12-58, -8%)	1 MHz, 20 C
Marble, dolomitic (USGS sample, New York)	7.7	1 kHz	[13]	–	1 kHz, dry; tan delta 0.0137; single sample
Marble, pink (USGS sample, Colorado)	11.7	1 kHz	[13]	–	1 kHz, dry; tan delta 0.0213; single sample
Solid Insulation / Polymers					
Polyethylene (LDPE)	2.3	60 Hz	[3]	2.3 (13-17, +0%)	
Polyethylene (MDPE)	2.3	60 Hz	[3]	2.3 (13-17, +0%)	
Polyethylene (HDPE)	2.35	60 Hz	[3]	2.3 (13-17, +2%)	
Cross-linked polyethylene (XLPE)	2.3	60 Hz	[3]	2.3 (13-17, +0%)	
EPR	3.25	60 Hz	[3]	–	dielectric constant 3.0-3.5 given at 1 kHz only (middle used)
Polypropylene	2.25	60 Hz	[3]	2.3 (13-17, -2%)	
PTFE	2	60 Hz	[3]	2.1 (13-17, -5%)	
Glass-reinforced polyester premix	6.3	60 Hz	[3]	–	
Thermoplastic polyester	3.55	60 Hz	[3]	–	dielectric constant 3.3-3.8 given at 100 Hz (middle used)
Polycarbonate	3.17	60 Hz	[3]	2.92 (13-17, +9%)	
Epoxy (mineral filler)	5	60 Hz	[3]	–	
Epoxy (silica filler)	3.85	60 Hz	[3]	–	
Silicone rubber	3.65	60 Hz	[3]	–	
Phenol-formaldehyde	5.3	60 Hz	[5]	–	60 Hz
Nylon 6,6	4	60 Hz	[5]	3.5 (13-17, +14%)	60 Hz
Polystyrene	2.6	60 Hz	[5]	2.6 (13-17, +0%)	60 Hz
Kapton HN polyimide film	3.4	1 kHz	[7]	–	1 kHz, 25 and 50 um film (3.5 for 75 and 125 um), ASTM D150
Delrin 150 acetal (POM-H)	3.7	60 Hz	[9]	3.8 (13-17, -3%)	60 Hz, 73 F, ASTM D150
Mylar polyester film (PET)	3.3	100 Hz	[10]	3.25 (13-17, +2%)	100 Hz, 25 C, read from Figure 6 (Mylar C, 25 um); about 3.28 at 1 kHz
Kapton FN 120FN616 (FEP-coated polyimide)	3.1	frequency not stated	[8]	–	frequency not stated; ASTM D150

Entry	ε_r	Frequency	Src.	CRC	Note
Kapton FN 150FN019 (FEP-coated polyimide)	2.7	frequency not stated	[8]	–	frequency not stated; ASTM D150
Kapton FN 250FN029 (FEP-coated polyimide)	3	frequency not stated	[8]	–	frequency not stated; ASTM D150
PVC, rigid (Trovidur PN)	3.2	frequency not stated	[12]	3.39 (13-17, -6%)	frequency not stated
Kraft paper, oil-impregnated	3.3	frequency not stated	[3]	–	frequency not stated; approximately 3.1-3.5 (middle used); impregnant about 2.2, dried cellulose 6.5-10
Natural rubber (polyisoprene)	2.6	1 kHz	[14]	–	1 kHz, 27 C; CRC gives typical values (2.5 at 1 MHz)
Neoprene (polychloroprene)	6.6	1 kHz	[14]	–	1 kHz, 25 C; CRC gives typical values (6.3 at 1 MHz)
Solid Insulation / Thin-Film Dielectrics					
SiO2 film	3.9	1 kHz	[3]	–	
Si3N4	12.7	1 kHz	[3]	4.2 (12-48, +202%)	
Ta2O5	28	1 kHz	[3]	24 (12-48, +17%)	
HfO2	35	1 kHz	[3]	–	
Circuit Board Laminates					
FR-4 epoxy-glass	4.35	1 MHz	[4]	–	
Polyfunctional FR-4	4.35	1 MHz	[4]	–	
Multifunctional FR-4	4.35	1 MHz	[4]	–	
Polyimide-glass	4	1 MHz	[4]	–	
BT epoxy	3.975	1 MHz	[4]	–	
Cyanate ester-glass	3.55	1 MHz	[4]	–	
PPO epoxy	4	1 MHz	[4]	–	
PTFE-glass	2.5	1 MHz	[4]	–	
Alumina substrate	8.9	1 MHz	[4]	–	
Semiconductors					
Silicon	11.7	static	[6]	12.1 (12-48, -3%)	300 K
Germanium	16.2	static	[6]	15.8 (12-48, +3%)	300 K
Gallium arsenide (GaAs)	12.9	static	[6]	13.13 (12-48, -2%)	static, 300 K
Indium phosphide (InP)	12.5	static	[6]	12.61 (12-48, -1%)	static, 300 K
Gallium nitride (GaN, wurtzite)	8.9	static	[6]	–	static, 300 K
Silicon carbide, 6H (c-axis along y)	9.66 / 10.03	static	[6]	9.66 (12-48, +0%)	static, 300 K; 9.66 perpendicular to c, 9.66/0.9631 = 10.03 parallel to c
Diamond	5.7	static	[6]	5.66 (12-48, +1%)	10^2-10^4 Hz

References

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[9] *Delrin 150 NC010 acetal (POM-H)*, technical data (ASTM D150).

[10] DuPont Teijin Films, *Mylar Electrical Properties*, Fig. 6 (values read from the graph).

[11] Corning, *Glass Silicon Constraint Substrates* (Pyrex 7740).

[12] Röchling, *Trovidur PN grey* (PVC-U), technical data sheet.

[13] G. R. Olhoeft, *Tables of Room Temperature Electrical Properties for Selected Rocks and Minerals with Dielectric Permittivity Statistics*, USGS Open-File Report 79-993, 1979, Table 2.

[14] W. M. Haynes, ed., *CRC Handbook of Chemistry and Physics*, 97th ed., CRC Press, 2016: Permittivity (Dielectric Constant) of Water at Various Frequencies (p. 6-15, from Archer and Wang, JPCRD 1983), of Liquids (6-199, C. Wohlfarth, from Landolt-Börnstein IV/6, 1991), of Gases (6-221, from NBS Circular 537 and Harvey and Lemmon 2005) and of Inorganic Solids (12-48, from Young and Frederikse, JPCRD 1973); Dielectric Constants of Glasses (12-58); Dielectric Constant of Selected Polymers (13-17, "typical or average values"). Source of the natural-rubber and neoprene entries; otherwise used as a check.

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